

Background:

Mycobacterium tuberculosis (Mtb) is the bacterium that causes tuberculosis. It attacks the lungs of the host it is in and causes granulomas to form when the immune system tries to destroy it. Figure 1 shows an example of this, as well as the way Mtb spreads. It is a serious problem, as it is one of the leading causes of mortality in the world today (Smith, 2003). Mtb protects itself from drug treatment by using toxin-antitoxin (TA) systems to shut down its processes, which turns it into a persister cell. Through this change, it evades antibiotic treatment, since the metabolism Mtb has is one of the factors that antibiotics need to target to destroy Mtb (Fisher et al., 2017). One of the TA systems that is of interest to analyze is VapBC35, which appears to be in the same operon as Mce3R, a regulator involved in cholesterol metabolism and stress resistance. This operon arrangement is unusual for a TA system, and the regulation of mce3R-vapBC35 is poorly defined. Mce3R binding sites may be an important asset in helping Mtb survive in a host. By finding out where Mce3R binds DNA, more of its functions and how it works would be revealed, which could lead to better treatment of tuberculosis. This can be done by finding the sequence motifs that Mce3R binds to, which I will cover here.

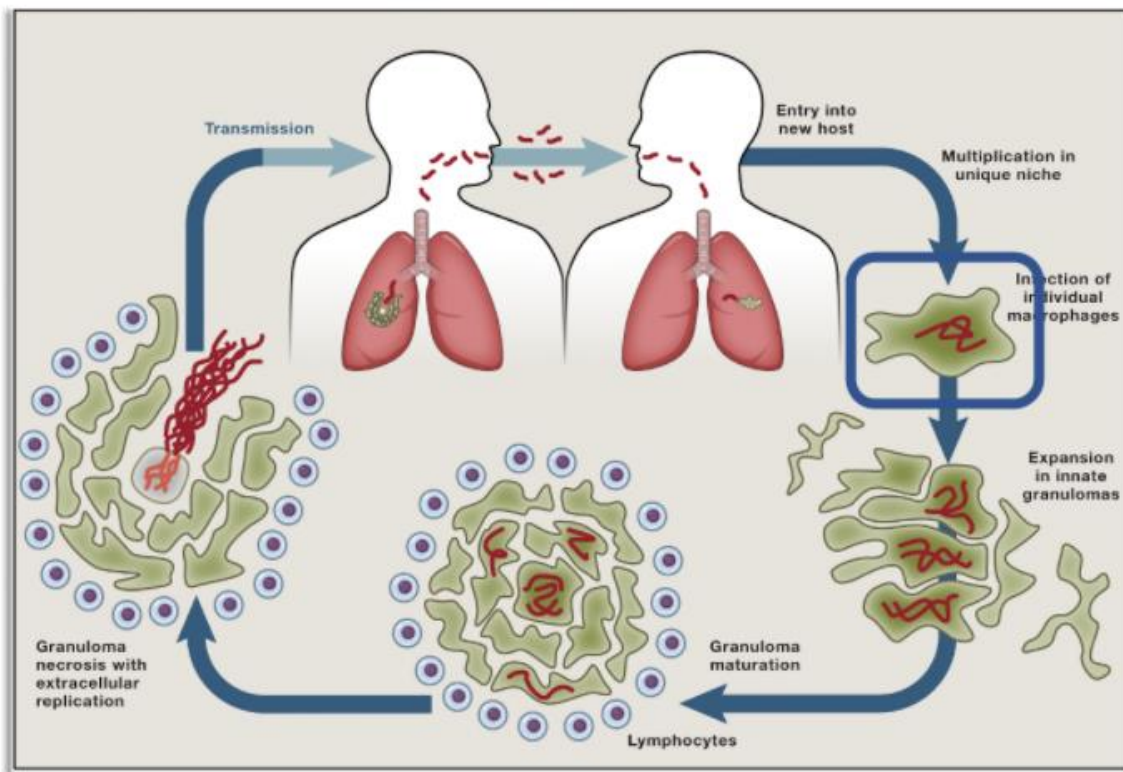


Figure 1: Mtb effects on lungs, as well as how it spreads. Taken from (Cambier et al., 2014).

Mycobrowser:

I used Mycobrowser to find gene locations of genes related to the intergenic regions of the mce3R regulon. For example, finding the intergenic region of mce3R-yrbe3a meant that I would have to find the locations of mce3R and yrbe3a. I found the locations of the genes by entering the gene names into the search bar of Mycobrowser. I only clicked on Mtb genes with no locus name underscores in the next page. The following page then gave the locations of the gene in the “Coordinates” box (only CDS locations were considered). I modified the gene locations to become the genomic region locations, based on the region name. In the same example, for mce3R-yrbe3a, the genomic region starting point would be (mce3R ending gene location + 1) and the ending location would be (yrbe3a starting gene location). I obtained the intergenic regions names through (Santangelo et al., 2009)’s data analysis. The isolated regions were

mce3R-yrbE3a (henceforth referred to as region 1), echA13-Rv1936 (region 2), Rv1941-Rv1942c (region 3), Mce3F-Rv1972 (region 4), and Rv1944c-Rv1945 (region 5). Figure 2 shows where each region is located. **Important Note:** Regions 3 and 4 are not possible regions that can hold Mce3R or VapBC35. I analyzed them to simply see what motifs they could hold, if any.

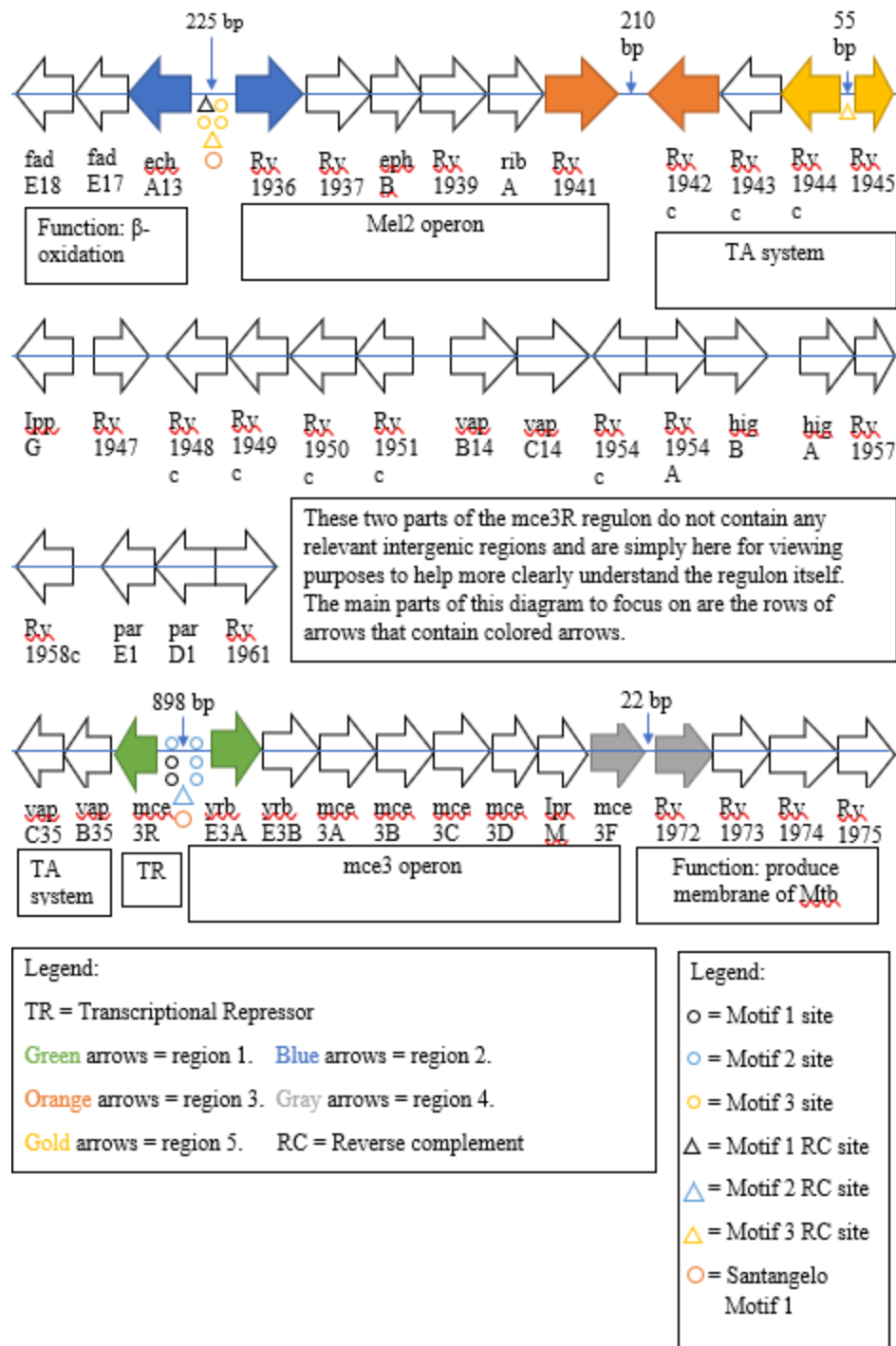


Figure 2: The *mce3R* regulon. The colored arrows represent the genes who share an intergenic region. The line between the colored arrows is the intergenic region of the genes mentioned underneath each colored arrow. The number of base pairs for each intergenic region is shown above the intergenic region. The direction of the arrows indicates which way the gene is transcribed (open reading frame). Three of the five intergenic regions (1, 2 and 5) have motifs. Santangelo's Motif 1 is located in Regions 1 and 2, the same places where my motif 1 is located.

TuberCuList:

I entered the obtained genomic region locations into the website “TubercuList,” which resulted in the extraction of DNA content (sequence data) of the intergenic regions. I input the locations of the genomic regions into the location boxes on the website and clicked enter. This led to a blue page popping up, in which I entered the region locations again at the near bottom of the page. I set the export settings to “View” for data extraction, I output direct gene sequences for the extraction, and Windows operating system was selected since I was using Windows. I clicked on “Get Data”, and I copy-pasted the sequences into text files using NotePad. Figure 3 shows an example of the sequence data extracted.

Important Note:

Regions 3 and 4 were not extracted from TuberCuList using this method. I extracted them by typing in the name of the first gene of each region (Rv1941 and mce3F, in these cases) into Mycobrowser’s search bar, clicking on the gene’s name, scrolling down to the Genomic sequence box, and typing in the number that corresponded to the DNA base amount between the genes of each region. In region 3’s case, the DNA base amount between Rv1941 and Rv1942c would be input. I always input the base DNA amount into the upstream flanking region box, regardless of what region was being extracted. I calculated the base amount by subtracting (ending point of the region’s first gene + 1) from (starting point of the region’s second gene).

```
>Seq1 [organism=Mycobacterium tuberculosis]
cggctcgaaccccccttcttgcgccaccggaatcacaaatcctgctagcagcatcgccctcag
cttcacccccgctcattcttcacctcgaatgcgccgggtcacccgggtgcgacacttacgggc
cgtcgtttcatgggtgacgtttcgaggctgtgctgctgccaaagaccccaggaagtctcggac
gagagactcgctagcctccgtgggtatcgggcatccctatcacccctgctcgatcctcaat
atcgggactaacaaaatacatcatcgcgccctgtatacgcgattacattgcaatttatcctt
atcaccccttcttagagtgcataatcagtaatagacatatcgcgctcctcgcgccccaggag
gcgggtcgacgaattcgccgtgcgcaacgacatgagccgtcgctgagcctgaaaacctgca
gacaaagcgcgagtgggggctggcaaaactacaggctcgtttagcagcaagttgcttcgac
gaccatgggtggcaacctcgccgggtcgcaaggctctgggtcggcggggccgaatcgaggcg
gtcaggatgcggcatccgatcacccgcccgtcgggcgcgctgttgatgcctgatcgtgggtg
cctcgccagcgtgactcgagccaacggcttgaccgggtgatgcgcctgtcggccgccaagg
cagcagagcacatcgccccgcgctataggatactagcaagatacatcatagccaatatat
gccagtttgcatgtattttaccgatcagttgtccaagcaatcgcgatttggtatggac
atcagcgggttctgccgcgtacgctcaccaatgtcacccgatcgtcgacctgtccggggggc
cagcgtgcgccacctcacccaacggcccagcatcgaatccagctgggtgcgccgcgcca
```

Figure 3: Example of sequence data extracted from TuberCuList (region 1).

Multiple Em of Motif Elicitation (M.E.M.E.):

Once the sequence data was extracted, I inserted the data into the website titled “M.E.M.E.,” which stood for Multiple Em of Motif Elicitation. I input the files individually, I set recognition mode of motifs to any number of repetitions (anr), and I input 100 motifs to be found (I placed a higher limit (x2 of previous limit) continuously if 100 motifs or more were found, until all motifs were found). Figure 4 shows the settings mentioned. I started the search by clicking on “Start Search”. Doing this generated sequence motifs of the data provided.

Data Submission Form

Perform motif discovery on DNA, RNA, protein or custom alphabet datasets.

Select the motif discovery mode ?

☒ Classic mode
 ☐ Discriminative mode
 ☐ Differential Enrichment mode **NEW**

Select the sequence alphabet

Use sequences with a standard alphabet or specify a custom alphabet. ?

☒ DNA, RNA or Protein
 ☐ Custom
 Choose File
No file chosen

Input the primary sequences

Enter sequences in which you want to find motifs. ?

Upload sequences
Choose File
intregion2_225.txt
DNA ?

Select the site distribution

How do you expect motif sites to be distributed in sequences? ?

Any Number of Repetitions (anr)

Select the number of motifs

How many motifs should MEME find? ?

100

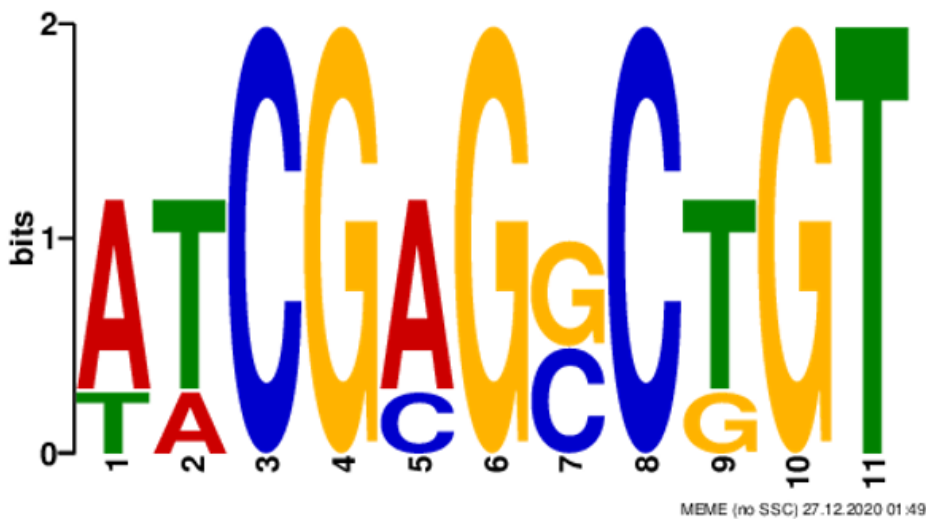
Figure 4: Settings used for motif generation on M.E.M.E.

Motif testing (Part 1):

I did multiple tests. The first test consisted of simply putting each region's sequence data in individually, which yielded multiple motifs for each of the 5 regions. I noted these motifs and saved them for later. Figure 5 shows an example of one of the motifs generated. The last test I would do would be more complex. The last test involved mixing the region sequence data together in multiple ways, and then finding motifs in those mixed sequences. To reduce testing, I did two tests before the last test, one to determine whether or not the reverse complement of sequences would affect combining sequences and one to know if positioning of sequences would affect combining sequences. I did the reverse complement test first by making a Python code to produce the reverse complement of one of the sequences given and inputting it into M.E.M.E. I found that the reverse complement sequence gave different motif results when compared to the regular sequence's motif results. That meant the reverse complement of the regions had to be analyzed. The next test done was the positioning of sequences test, to determine whether or not a

combination region would have the same motifs as the same combination region but with different sequence ordering (ex. Would region 1 + region 2 have the same motifs as region 2 + region 1?). To do this, I chose regions 1 and 2 as the regions to combine. I first made the combination of region 1+region 2, where region 1's sequence was preceding region 2's sequence, and tested it on M.E.M.E. Then, I made the combination of region 2+region 1, where region 2's sequence was preceding region 1's sequence, and tested it on M.E.M.E. I found that the motifs generated were different. Therefore, both reverse complements and sequence positioning of regions had to be considered for testing. With these two tests done, I proceeded with the making the actual region combinations for the last test.

E-value: 1.7e+004 [?](#) **Site Count:** 4 [?](#) **Width:** 11 [?](#)



Log Likelihood Ratio: 48 [?](#) **Information Content:** 17.8 [?](#) **Relative Entropy:** 17.5 [?](#) **Bayes Threshold:** 7.52193 [?](#)

Name ?	Strand ?	Start ?	p-value ?	Sites ?
1. Seq1	-	451	1.43e-6	ACTTGCTGCT AACGAGCCTGT AGTTTTGCCA
1. Seq1	+	139	1.43e-6	atggtgacgt TTCGAGGCTGT gctgctgcca
1. Seq1	+	532	2.80e-6	gcgggcccga ATCGAGGCGGT caggatgcgg
1. Seq1	+	262	2.80e-6	aaaatacatc ATCGCGCCTGT atacgcgatt

Figure 5: Example of motif generation done in M.E.M.E.

Motif testing (Part 2):

I determined how many possible combinations of regions there would be by using simplistic math. Since there were 5 regions, that meant there would be 2-region combinations, 3-region combinations, 4-region combinations, and 5-region combinations. I used the combination formula to find all sequence combinations. For 2-region combinations, the total number of combinations was $\frac{5!}{2!(5-2)!} = 10$. For 3-region combinations, it was $\frac{5!}{3!(5-3)!} = 10$. For 4-region combinations, it was $\frac{5!}{4!(5-4)!} = 5$. For 5-region combinations, it was $\frac{5!}{5!(5-5)!} = 1$. In total, I needed to test 26 regions. However, since reverse complements had to be considered for all regions, as well as sequence positioning, I needed to test at least 662 combination regions. However, upon further testing, I found that a majority of regions, when comparing their reverse complement and regular forms, had no significant motif changes between forms, meaning that all motifs in the reverse complement form matched the motifs of the regular form. I also found that positional changes yielded the same results. I checked the results of the last two tests to see if I misinterpreted them. I found that the significant motif changes were actually non-existent: the motifs between forms were the same, just with slight alterations (extra letters added or removed). I also found that the slightly altered motifs appeared a lot less than or had higher E-values than the originals, making them not superior to the original motifs. This meant that these altered motifs are not relevant enough to analyze. As a result, I concluded that reverse complement sequences and positionally different sequences were found to not be worth analyzing. The amount of regions to be analyzed thus dropped to 26. I found all sequence combinations through combination trees that were made of every single possible combination. I only considered combinations valid if they differed by more than just order from already validated combinations. If they only differed by order, I did not consider them. Using this validation method for each

option, I found all combination regions. I then did the last test, with each combination region being input into M.E.M.E. I noted all motifs. I then looked at every single motif from the first test (individual regions) and the last test (combination regions) and only considered motifs with low E-values. These motifs are listed as figures 6-8 below.

Results:

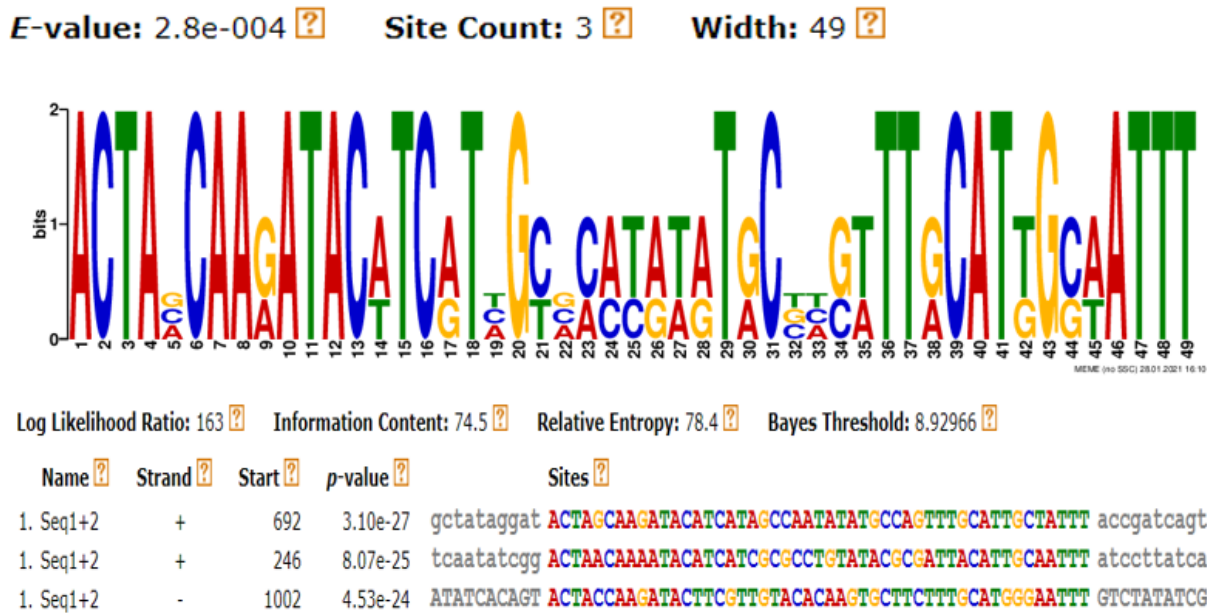
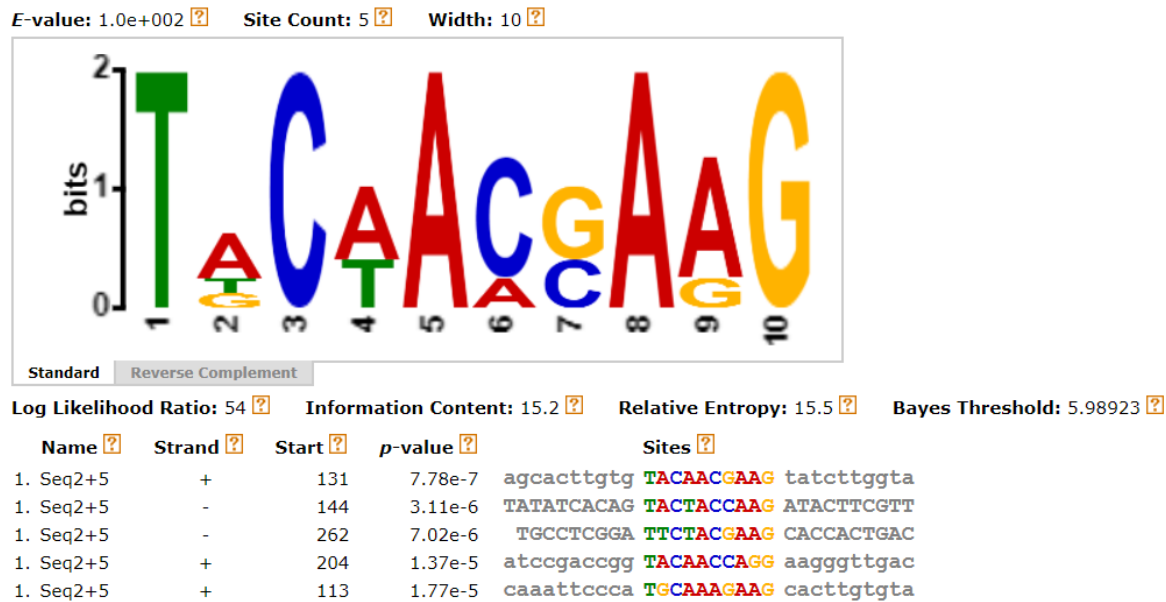
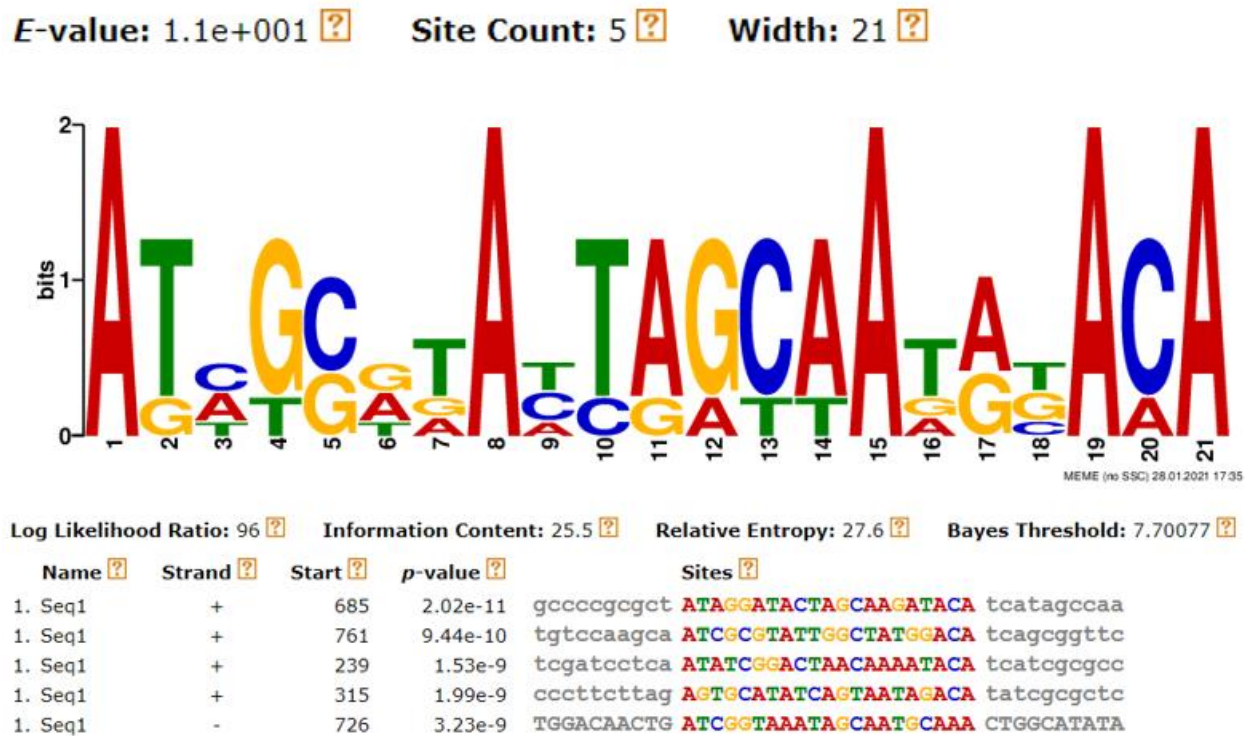


Figure 6: Most likely motif candidate for Mce3R/VapBC35 binding.



Figures 7-8: Other potential candidates for Mce3R/VapBC35 binding, in order of descending importance (Figure 7 is more important than Figure 8).

I consider these the most important motifs for analysis, as they have low E/P values compared to all other motifs found (arranged in order of descending importance). The fact that these motifs have low E/P values would strongly suggest that they could be binding sites for mce3R or VapBC35. In addition, all of the sequences have reverse complement locations, suggesting they can be VapBC35 binding sites as well as mce3R binding sites.

The intergenic regions where the motifs were found are shown on the next page within Figures 11-13.

>Regions1+2 [organism=Mycobacterium tuberculosis]

cggctcgaaccccttcttgcgaccggaatcacaatcctgctagcagcatcgccctcagcttcaccccgtcattcttcacctcgaatgcgc
 cggtcaccgggtgcgacactaccggccgtcggtcatggtgacgttcgaggctgtgctgctgccaagaccccaggaagtctcggacgag
 agactcgtagcctccgtggtatcgggcatccctatcacccctgctcgatcctcaatatcggaactaacaaaatacatcatcgcgctgtatac
gcgattacattgcaatttatccttatcaccccttcttagagtgcataatagacatatcgcgctcctcgcgccccaggaggcggtcgacg
 aattcgccgtgcgcaacgacatgagccgtcgctgagcctgaaaacctgcagacaaagcgcgagtggggggtggcaaaactacaggctc
 gttagcagcaagttgcttcgacgacctgggtggcaacctcgccggtcgcgaaggctctggtcggcgggcccgaatcaggcggtcagga
 tgcggcatccgatcaccgcccgtcgggcgcgctgttgatgcctgatcggtgcctcgccagcgtgactcgagccaacggcttgaccggt
 gatgcgctgtcggccgccaaggcagcagagcacatcgccccgcgtataggatactagcaagatacatcatagccaatatatgccagttt
gcattgctatttaccgatcagttgtccaagcaatcgcgatttgctatggacatcagcggttctgccggtacgctcaccaatgtcaccgatcg
 tcgacctgtccggggggccagcgtgcgccacctacccaacggcccagcatcgaatccagctggtgcgccgcgcca█gtccgctcctta
 ctggacgtgaacggctgacctgtgcgaggttaccgctcccttagccaacatgtccatagccaatacgcacatgagagtgatcgatatagaca█
aattcccatgcaaagaagcacttgtgtacaacgaagtatcttggtagtactgtgatatacgcaaagggcgccaccgcagcgcgccgggcat
 ccgaccggtacaaccaggaagggttgacga

Legend:

Blue text = Figure 11 motif found in regular order

Red text = Figure 11 motif found in reverse complement order

Green text = starting position of regular order motifs

Pink text = starting position of reverse complement order motifs

█ = border between regions 1 and 2

Figure 11: Intergenic region for the motif most likely to be an Mce3R/VapBC35 binding site. This intergenic region shown here was a combination of regions 1 and 2. The arrows in the figure point to the starting positions of the motifs, and the captions next to each arrow show what numerical position the motif is at within the intergenic region (e.g. 246 means the motif starts at the 246th letter of the sequence). Santangelo motifs are underlined as follows: motif #1, motif #2, rev motif #1, rev motif #2.

>Region1 [organism=Mycobacterium tuberculosis]

```

cggctcgaaccccccttcttgcgcaccggaatcacaaatcctgctagcagcatcgccctcagcttcaccccgtcattcttcacctcgaatgcgc
cggtcaccgggtgacgacattaccggccgtcggtcatgggtgacgtttcgaggctgtgctgctgccaagaccccaggaagtctcggac
gagagactcgtagcctccgtggtatcgggcatccctatcacccctgctcgatcctcaatatcggactaacaaaatacatcatcgcgctgta
Position: 239
Position: 315
tagcgattacattgcaatttatccttatcaccccttcttagagtcataatcagtaataagacatatcgcgctcctcgcgccccaggag
gcggtcgacgaattcgccgtgcgcaacgacatgagccgtcgctgagcctgaaaacctgcagacaaagcgcgagtgggggctggcaaaa
ctacaggctcgtagcagcaagtgcttcgacgacctggtggcaacctcgccgggtcgcggaaggctctggtcggcgggcccgaatcgag
gcggtcaggatgcggcatccgatcacgcccgtcgggcgcgctgttgatgctgatcggtgcctcgccagcgtgactcgagccaacg
gcttgaccggtgatgcgctgtcggccgccaaggcagcagagcacatcgccccgcgtataggatactagcaagatacatcatagccaat
atagccagttgcattgctatttacgatcagttgtccaagcaatatcgctattggctatggacatcagcggttctgccggtacgctcaccaat
Position: 726 Position: 761
gtcaccgatcgtcgacctgtccggggggccagcgtgcgccacctcaccaacggcccagcatgaatccagctggtgcgccgcgcca

```

Legend:

Blue text = Figure 12 motif found in regular order

Red text = Figure 12 motif found in reverse complement order

Green text = starting position of regular order motifs

Pink text = starting position of reverse complement order motifs

Figure 12: Intergenic region for the motif that is the second most likely to be an Mce3R/VapBC35 binding site. This intergenic region shown here was simply region 1. Santangelo motifs are underlined.

>Regions2+5 [organism=Mycobacterium tuberculosis]

gtccgctccttactggacgtgaacggctgacctgtgcgaggttacccgtcccttagccaacatgtccatagccaatacgcacatgagagtga
 Position: 113 → Position: 131 →
 tcgatatagacaaattcccatgcaagaagcacttggtacaacgaagtattcttggtagtactgtgatatacgcaaagggcgccaccgcagc
 Position: 144 →
 gcgccgggcatccgaccggtacaaccagggaagggttgacga■tccgagcctatgacggcccgggtctgtcagtggtgcttcgtagaatc
 cgaggca Position: 204 Position: 262

Legend:

Blue text = Figure 13 motif found in regular order

Red text = Figure 13 motif found in reverse complement order

Green text = starting position of regular order motifs

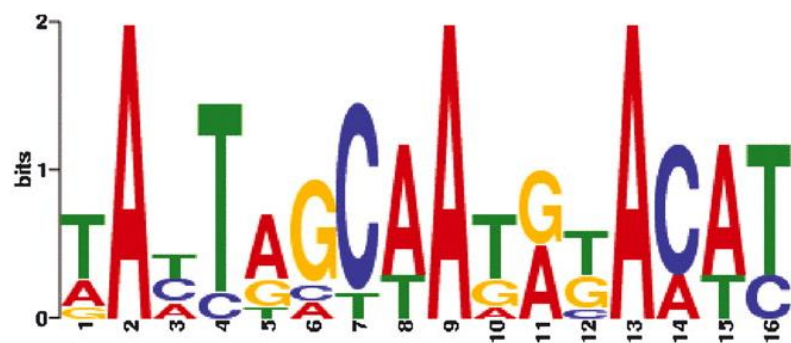
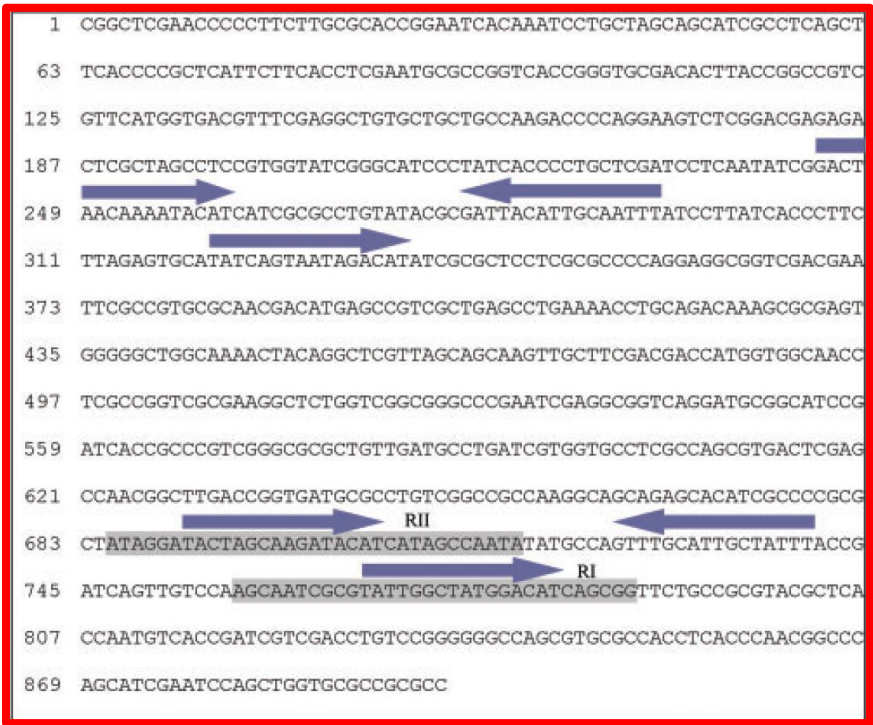
Pink text = starting position of reverse complement order motifs

■ = border between regions 2 and 5

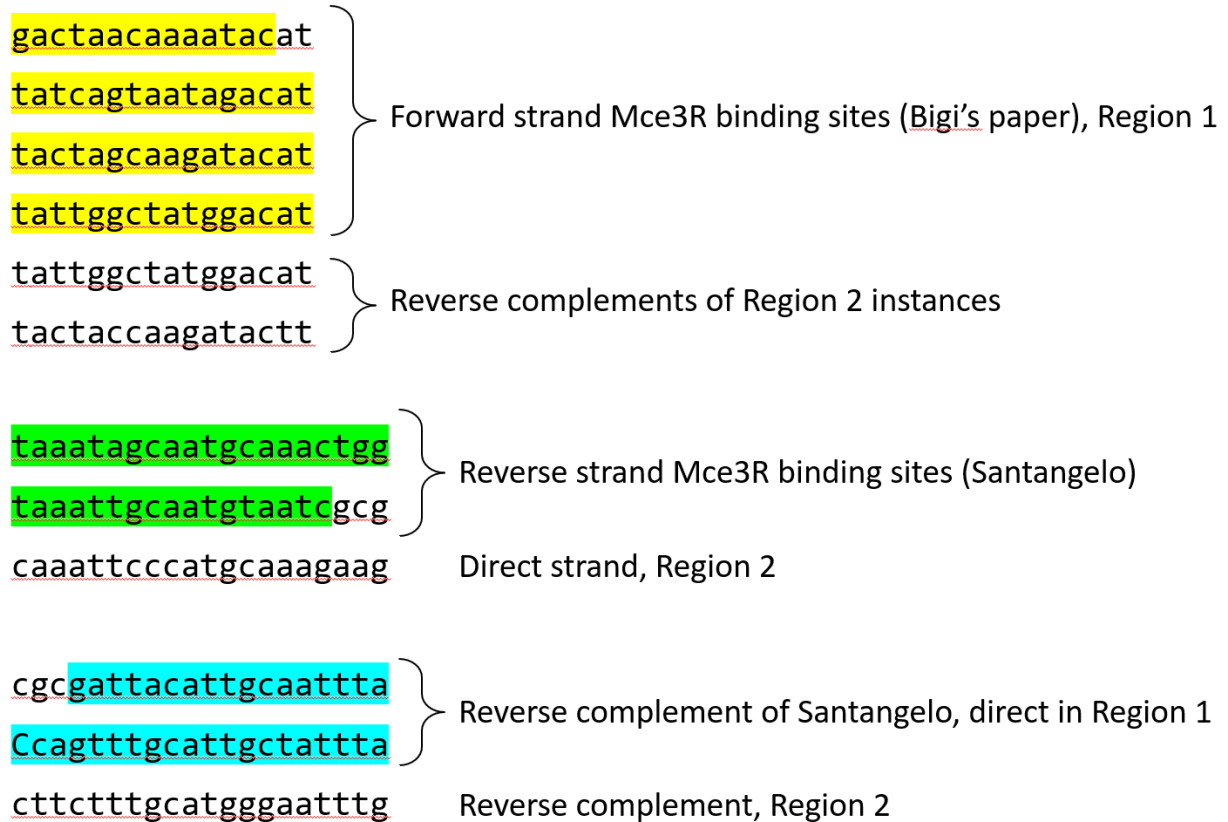
Figure 13: Intergenic region for the motif that is the least likely to be an Mce3R/VapBC35 binding site. This intergenic region shown here was a combination of regions 2 and 5.

Appendix:

Motifs from Santangelo et al., Microbiology 155(Pt 7):2245-2255 (2009).



There seem to be multiple issues with the motifs annotated in 2009. They have different lengths, and unmatching sequences.



References:

Cambier, C. J., Falkow, S., & Ramakrishnan, L. (2014). Host evasion and exploitation schemes of *Mycobacterium tuberculosis*. *Cell*, 159(7), 1497–1509.

<https://doi.org/10.1016/j.cell.2014.11.024>

Fisher, R. A., Gollan, B., & Helaine, S. (2017). Persistent bacterial infections and persister cells. *Nature reviews. Microbiology*, 15(8), 453–464.

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Smith I. (2003). *Mycobacterium tuberculosis* pathogenesis and molecular determinants of virulence. *Clinical microbiology reviews*, 16(3), 463–496.

<https://doi.org/10.1128/cmr.16.3.463-496.2003>